

Nick's Question 0.1 Details from Remark 1.1 page 3 that part (i) implies the existence of K .

Is this correct in showing

Let $\psi \in C_0^\infty(\mathbb{R}^d)$ then we have

$$\begin{aligned} \int_{\mathbb{R}^d} dx \gamma(x) \psi(x) &= \int_{\mathbb{R}^d} dx \varphi(x) \mathcal{F}(\psi)(x) \\ &= \int_{\mathbb{R}^d} dx \mathcal{F}(\varphi)(x) \psi(x) \\ &= \int_{\mathbb{R}^d} dx \mathcal{F}(\sqrt{\varphi} \sqrt{\varphi})(x) \psi(x) \\ &= \int_{\mathbb{R}^d} dx \left(\mathcal{F}(\sqrt{\varphi}) * \mathcal{F}(\sqrt{\varphi}) \right)(x) \psi(x) \end{aligned}$$

and since this is true for all test functions ψ then we must have $\gamma(x) = \left(\mathcal{F}(\sqrt{\varphi}) * \mathcal{F}(\sqrt{\varphi}) \right)(x)$

Nick's Question 0.2 top of page 6

Why does assumption A imply that $\alpha \in (0, 2 \wedge d)$?